Lesson 5-4 · Period 3

Radical Equations — Assessment-Critical: Isolate a Variable + Table Completion

Klimsara-adapted · Student-centered · No Blooket

Do Now — Extraneous Solutions (HOT)

DOK 3 | 5 min | *bridge from P2*

Higher Order Thinking · Extraneous Solutions Some, but not all, equations with rational exponents have extraneous solutions.

Think: What is the relationship between the exponents and the possibility of having extraneous solutions? Explain your reasoning.

Predict: If you solved $V = \frac{1}{3}\pi r^2 h$ for r, would you expect an extraneous solution? Why or why not?

Launch — Rewrite a Formula + Table Completion

DOK 2 | 12 min | *teacher models Ex 2 + #44*

Isolate-a-Variable Rules

1. Identify which variable to isolate.
2. Undo operations LAST-IN, FIRST-OUT: add/sub → mult/div → exponent/radical.
3. Raise both sides to the **RECIPROCAL** power to undo $x^{m/n}$: flip to $x^{n/m}$.

Table-Evaluation Rules

1. Substitute each input into the rewritten formula.
2. Simplify rational exponents step-by-step (use calculator for irrational outputs).
3. Round to specified precision.

Launch — Ex 2 (Golden Gate) + Practice #44

DOK 2 | 12 min | *teacher models*

Example 2 · Rewrite the Golden Gate formula Given: $x = \dfrac{\sqrt{y - 220}}{0.010583}$ Isolate y.

Steps: multiply both sides by coefficient $\rightarrow$ square both sides $\rightarrow$ add 220.
Then substitute $x = 200$ and evaluate y.

Practice #44 · Table Completion **ASSESSMENT REHEARSAL — Item 2**

Complete the table for $y = \sqrt[3]{2x + z} - 12$. Teacher walks through rows 1–2 live. Copy the steps; note calculator workflow for irrational outputs.

Explore — Think-Pair-Share (35 min)

DOK 2 | 35 min | *student work — TPS*

6 items in order. Never cut #10 (Item 1A rehearsal). Cut #27 first if pace slows.

Try It 2 · Stopping distance formula $v = 3.57\sqrt{d}$ Rewrite to isolate d.

Practice #10 · Isolate x ASSESSMENT ITEM 1A REHEARSAL Rewrite $y = \sqrt{(x-48)/6}$ to isolate x.

Practice #25 / #26 / #27 · Rationalization pattern #25: $x = 3\sqrt[3]{15+y}$ #26: $x = \frac{\sqrt{2y}}{26}$ #27: $x = \frac{\sqrt{y-14.2}}{0.05}$ Solve for y.

DOK 2 **5 min** *concept summary + assessment preview*

Sentence Frame (#10 or #39) "To isolate ____, I first ______, then raise both sides to the power ____. The rewritten formula is ______. Plugging in ___ = ___ gives ____. In context this means _______."

Topic 5 Assessment Preview **Items 1A/1B** — Rewrite a pyramid-volume formula for x, then evaluate. ⇒ Rehearsed by **Practice #10**.

Item 2 — Complete the table from the rewritten formula. ⇒ Rehearsed by **Practice #44** (done in Launch).

Exit Ticket — Pre-Assessment Confidence Check

Algebra 2 · Lesson 5-4 · Period 3

DOK 1-2 | 3 min | *not graded*

Complete on your exit ticket

1. To isolate a variable from a rational-exponent formula I ________________________.
2. When I evaluate the formula at a specific x, I ________________________.
3. One biggest thing I learned today: ___________________________.